THE LAWS OF SOFTWARE TEAMS

The Peter principle

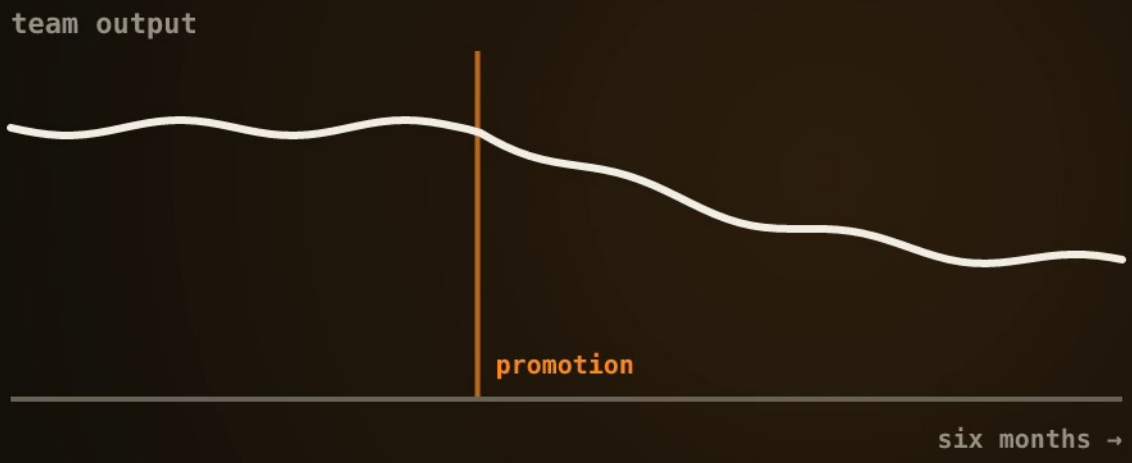
In a hierarchy, people rise to their level of incompetence

swipe →

Ivan Rykov

Best engineer, new manager



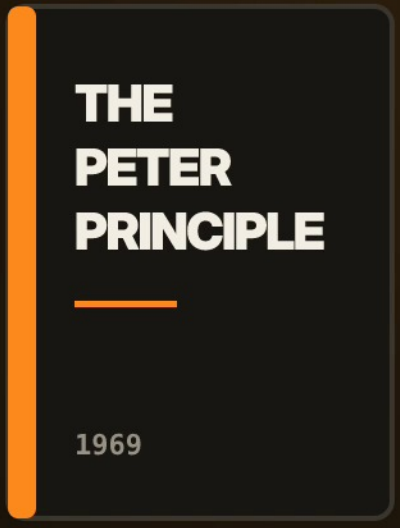


A slower team, and one engineer fewer

This has a name

illustration

“In a hierarchy, every employee tends to rise to his level of incompetence”



Laurence J. Peter
& Raymond Hull

William Morrow, 1969

written as satire
and read as truth

Promoted for the job they already do



Each promotion rewards the last job

illustration

Satire, but the data agrees

53,035 sales workers

214 firms · 2005-2011



1 dot ≈ 500 workers

1,531 promoted to manager

BENSON, LI & SHUE · 2018

NBER working paper

Top sellers made weaker bosses

DOUBLE YOUR SALES

+14.3%

chance of promotion



DOUBLE THE NEW MANAGER'S OLD SALES

−7.5%

sales per team member



A BETTER SIGNAL

Shared sales

with colleagues before the promotion

relative changes when sales double · an association, not proof
Benson, Li & Shue, 2018

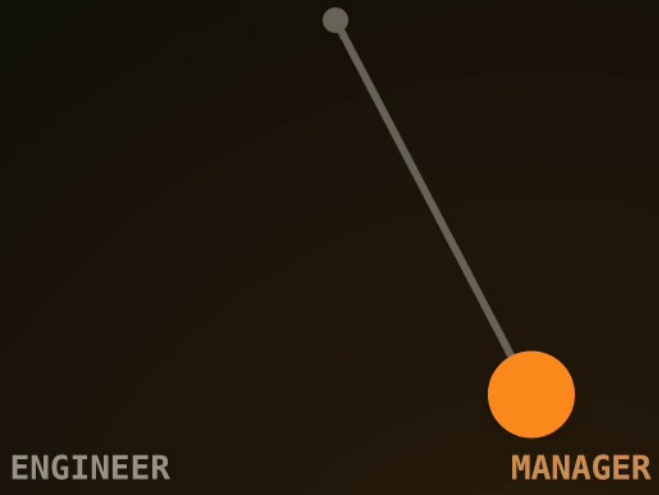
Two ladders, not one

TECHNICAL	MANAGEMENT
Principal	Director
Staff	Manager
Senior	Team lead
Engineer	Engineer

Try first: tech lead · lead one project

illustration · level names vary by company

Make the way back safe



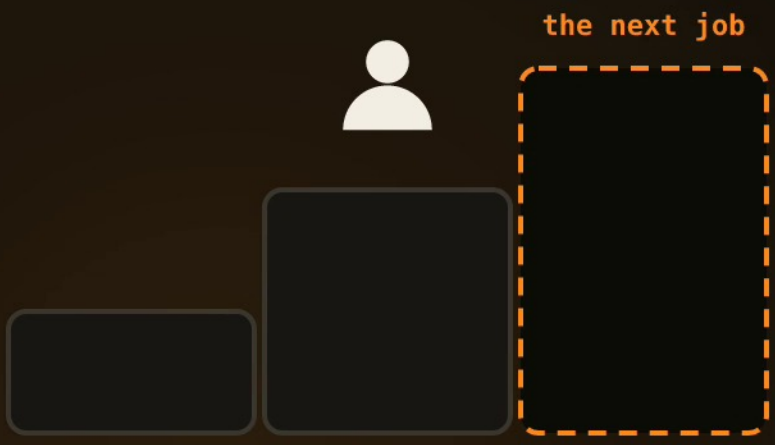
CHARITY MAJORS · THE ENGINEER/MANAGER PENDULUM · 2017

> “Management is not a promotion, management is a change of profession”

charity.wtf, 11 May 2017

Going back is not a failure

Promote for the next job



Not for the last one

WHAT I TAKE FROM THIS

This is how I choose and grow leads:

1. **The most important rule: I do not give a role to someone only because they worked next to it. Doing the routine work around a role is not the same as driving it. I need a person who covers the area with their head, not only with their hands. They decide what to do, say what they think and push the work forward. Routine work can often be automated, but an area needs someone who thinks about it**
2. **A people lead needs empathy and thinks about the product, not only about the next task. They see the wider picture and take responsibility for it. But too much perfectionism can burn out the lead and the team. The work of the team matters more than a perfect result. A tech lead makes decisions, knows how to simplify, takes responsibility and does not panic when things move fast**

WHAT I TAKE FROM THIS

3. I do not promote out of friendship or as a reward. I also do not promote because there is no other choice: in my experience, such promotions fail most often
4. People can try before the title. One engineer who used to refuse leadership agreed to guide two engineers technically. Then came three, then four, and later this engineer led a key team. We went through this engineer's tasks and feedback together. With every new lead, I have weekly 1:1s at first and later every two weeks

WHAT I TAKE FROM THIS

5. **Leading people is not the only way up. A strong engineer can grow as a tech lead, join an architecture council or get more influence on decisions. Not wanting to lead people is normal. I offer a try when I see that a person can do it, even if they are not sure yet, and I do not push. Even after a try, some people do not feel ready, and that is normal too**
6. **I look at a new lead's team more than at technical results: how the work moves and what feedback people give. A good sign is a team that works together smoothly. Warning signs are anger at the team, not understanding what is going on, overwork and burnout**

WHAT I TAKE FROM THIS

7. **The way back must be safe. I have seen strong, responsible engineers become team leads, and their teams fell apart. A return to engineering is not a failure**

AI agents give a strong engineer some practice in leading: setting clear goals and limits, delegating, not being afraid to let go, checking part of the result and owning it. But an agent does not get offended and accepts any wording. People need empathy and an individual approach, and agents do not teach that

Sources: Peter and Hull, 1969 · Benson, Li and Shue, NBER 2018 · Charity Majors, 2017

next: Gilb's law